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39. C-SCAN Disk Scheduling

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main() {    int n, head,
max, seek = 0;    int a[20];

    printf("Enter number of requests: ");
    scanf("%d", &n);

    printf("Enter requests: ");
    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter initial head: ");
    scanf("%d", &head);

    printf("Enter disk size: ");
    scanf("%d", &max);

    printf("Head movement: %d", head);

    /* Move towards higher tracks */
    for(int i = 0; i < max; i++)
    {
        for(int j = 0; j < n;
j++)
        {
            if(a[j] == i && i >= head)
            {
                seek +=
abs(head - i);        head = i;
                printf("-> %d", head);
            }
        }
    }
}
```

```

    /* Move to end of disk */
    seek += abs(head - (max - 1));
    head = max - 1;    printf(" ->
%d", head);

    /* Reverse direction */
    for(int i = max - 1; i >= 0; i--)
    {
        for(int j = 0; j < n;
j++)
        {
            if(a[j] == i && i < head)
            {
                seek +=
abs(head - i);        head = i;
printf(" -> %d", head);
            }
        }
    }

    printf("\nTotal Seek Time = %d\n", seek);
    printf("Average Seek Time = %.2f\n", (float)seek / n);

    return 0;
}

```

Output:

```

Enter number of requests: 8
Enter requests: 77 18 45 33 12 43 123 88
Enter initial head: 53
Head movement: 53 -> 77 -> 88 -> 123 -> 0 -> 12 -> 18 -> 33 -> 43 -> 45
Total Seek Time = 238
Average Seek Time = 29.75

...Program finished with exit code 0
Press ENTER to exit console.

```